

Upom Bipro

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<https://github.com/upombiprocodes>

KEY SKILLS

Technical Skills

- **Programming & Data** – Python, Java, Javascript, SQL, scikit-learn, Streamlit; built full-stack apps and ML dashboards.
- **UI/UX & Databases** – Designed secure SQL schemas (1,000+ records) and Figma prototypes influencing project UX decisions.

Soft Skills

- **Communication & Support** – Guided prospective students as TAP Ambassador and provided 1-to-1 SEND support as Teaching Assistant.
- **Teamwork & Problem-Solving** – Collaborated with teachers, parents, and developers to automate processes and meet diverse needs.
- **Organisation & Customer Service** – Balanced studies, technical projects and part-time roles while delivering high-quality service at M&S.

EDUCATION

Brunel University of London, Uxbridge

Sept 2024 - June 2028

BSc Computer Science (Predicted 1.1)

- **Year Two Modules:** Software Development and Management, Usability Engineering, Algorithms and their Applications, Networks and Operating Systems
- **Year One Modules (1.1 overall):** Introductory Programming (C), Information Systems and Organization (A), Logic and Computation (A+), Data and Information (A), Programming Application (A), Group Project (A), Interactive Learning Skills and Communication B (A).

Birshrestha Noor Mohammad Public College, Bangladesh

Jan 2009 - November 2023

- HSC (Physics, Chemistry, Higher Mathematics, ICT - All A*) | January 2022 - November 2023
- SSC (Physics, Chemistry, Mathematics, Information Technology) - All A*

TECHNICAL EXPERIENCE

Brunel University London Group Project – Group Member – United Kingdom

Sept 2024-May 2025

- Handled huge sets of databases using SQLite.
- Designed and developed an entire interactive management application using Figma and Java Swing.
- Integrated the software to a university system for a completely usable university portal.

Creative IT – Artificial Intelligence Engineer Intern – Dhaka, Bangladesh

Jun 2022-Dec 2022

- Worked on a team project in building and training ML models to predict traffic patterns with high accuracy.
- Conducted data collection and cleansing to improved dataset quality by noticeable margin.
- Contributed to a 20% improvement in model accuracy.

Amra IT – Frontend Junior Web Developer – Dhaka, Bangladesh

Jan 2021-Mar 2022

- Developed responsive websites by designing using Figma, optimising for user experience and SEO.
- Implemented interactive JavaScript and CSS features to enhance usability.
- Associated with clients regularly to meet project requirements and deadlines.

TECHNICAL PROJECTS

Traffic Prediction System (*Python, scikit-learn, pandas, SHAP, Streamlit, Folium, joblib, DBSCAN*)

[GitHub – upombiprocodes/Traffic-Prediction-System](#)

Individual Project (May 2025 - Aug 2025)

- **Built an end-to-end machine learning platform** in **Python** to predict hourly traffic volumes from 10+ environmental and location-based features using a **RandomForestRegressor** model (trained with **scikit-learn** and **pandas**).
- Engineered Multiple Modules:
 - Anomaly detection with **IsolationForest**
 - Hotspot detection using **DBSCAN** and map visualisation with **Folium**
 - Event/incident feature enrichment and travel time suggestion logic.

- **Developed an interactive dashboard** in **Streamlit** with **SHAP** explainability plots to display real-time predictions and feature importance, improving interpretability of model decisions.
- **Automated data pipelines**: ingestion of CSV data (thousands of traffic records), model training/validation, and persistence with **joblib**; achieved a validation MAE on the full dataset.
- **Visualised high-traffic hotspots** on interactive maps to support congestion analysis and planning; reduced manual analysis time by **inserting percentage reduction** compared with previous ad hoc approaches.

Applicant Management App (Figma, Java, SQL)

<https://github.com/upombiprocodes/Applicant-Management-System-UNIVERSITY>

Personal Module of Group Project (Oct 2024- May 2025)

- Engineered a university Applicant Management System, achieving over 90% automation of manual admissions processes and reducing administrative workload.
- Designed and implemented SQL database schemas, managing 100% of applicant data securely and enabling efficient data retrieval for over 1,000+ records.
- Prototyped intuitive user interfaces in Figma, directly influencing 80% of UI/UX design decisions and improving user satisfaction scores.
- Developed and maintained backend logic with Java Swing, supporting 100% of core application workflows and ensuring seamless integration with the database.
- Demonstrated full-stack development by combining SQL data handling, Figma UI design, and Java Swing backend, covering 100% of the project lifecycle from concept to deployment.

PART-TIME EMPLOYMENT

Retail Assistant – M&S

Nov 2025-Ongoing

- **Customer Service**: Assist customers by answering questions and helping them find products.
- **Stock Management**: Organise and replenish inventory on shelves and in storage.
- **Checkout Operations**: Operate cash registers and handle transactions accurately.

TAP Ambassador – Brunel Pathway College

Mar 2025 - Ongoing

- **Promoted Brunel Pathway College's programs and student life** to prospective students through The Ambassador Platform (TAP), increasing engagement and conversions.
- **Provided personalised guidance and answered queries** from prospective students, supporting them with informed decisions about higher education.
- **Created and shared authentic student experiences** via posts, messages, and events to strengthen the university's outreach and online presence.

Teaching Assistant – AXCIS Education London

Jan 2025 – Ongoing

- **Provided tailored 1-to-1 and small-group support** to pupils with Special Educational Needs and Disabilities (SEND), helping them access the curriculum and achieve learning goals.
- **Implemented behaviour and learning strategies** under teacher guidance, fostering a safe, inclusive, and supportive classroom environment.
- **Teamed up with teachers, parents and external professionals** to track progress, adapt resources, and ensure consistent support for each child's individual education plan (IEP).

ACHIEVEMENTS

- Student Ambassador, Brunel Plus Gold Award (predicted), Brunel Plus Silver Award, First Aid Certification, SIA Door Supervisor License Holder.

INTERESTS

- Brunel Blockchain Society, Brunel Tech Society, Coding, Puzzle Solving, Football, Debating, Competitive Games, Travelling.

LANGUAGES

- English (C1), Bangla (Native), Hindi (Fluent), Urdu (Fluent)